

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

Anonymous authors
Anonymous submission

ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has become the dominant paradigm for advancing the reasoning capabilities of large language models. Yet the prevailing family of algorithms typified by GRPO is systematically undermined by vanishing gradients, a failure mode we trace to two coupled and mutually reinforcing design choices: the fixed clipping bounds imposed on token-level probability ratios, and the step-local standardization of identical rewards. Acting together, these choices produce ineffective gradient updates and leave a large fraction of generated responses entirely unexploited. We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a unified policy-optimization method that resolves both bottlenecks at their source through two complementary and jointly designed mechanisms. First, a dynamic-adaptive clipping strategy adaptively widens the admissible update interval for each token as a monotone function of its prior probability, restoring token-level exploration exactly where it is most informative. Second, a smooth advantage standardization procedure aggregates reward statistics across cumulative training steps, substantially improving the response-level utilization of generated data. DCPO establishes state-of-the-art performance across four mathematical reasoning benchmarks and four distinct models spanning both instruct and base variants. On the AIME24 benchmark with Qwen2.5-Math-7B, DCPO attains an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-fold sampling, decisively surpassing DAPO (36.7/31.6), GRPO (36.7/32.1) and GSPO (40.0/34.9); on the AIME25 benchmark with Qwen2.5-14B, DCPO reaches (23.3/19.0), again outperforming GRPO (13.3/10.5), DAPO (20.0/15.3) and GSPO (16.7/9.9). These accuracy gains are inseparable from concomitant efficiency gains: DCPO raises the nonzero-advantage ratio by an average of 28% over GRPO across the four models, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude relative to both GRPO and DAPO, all while delivering superior end-task performance. Collectively, these results establish that DCPO converts generated data into usable learning signal markedly more efficiently, providing a general and effective foundation for reinforcement learning in large language models.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central focus of contemporary research (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Among the techniques driving this progress, Reinforcement Learning from Verifiable Rewards (RLVR) has established itself as the dominant paradigm: by optimizing LLMs against rule-based outcome rewards, RLVR delivers its most pronounced gains precisely in the domains that demand reflective, multi-step decision-making, such as mathematics and coding (DeepSeek-AI et al., 2024).

The GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms was developed to operationalize RLVR at scale, and although its strong empirical performance has been validated across numerous studies, recent work such as DAPO (Yu et al., 2025) has exposed two critical and mutually reinforcing limitations that constrain it. First, GRPO is prone to entropy collapse: as advantage values decay toward zero during the later stages of training, the learning signal is progressively silenced. Second, its indiscriminate weight-clipping discards rarely sampled tokens—precisely the signals that, by virtue of their low probability, carry the most information—thereby curtailing the diversity of model exploration. These two failure modes share a common root in the step-local, fixed-bound design of the underlying objective.

Existing remedies address these symptoms only partially. DAPO, for instance, introduces the Clip-Higher strategy to counter entropy collapse by relaxing the upper clipping bound, together with a Dynamic Sampling strategy that filters out prompts with uniform rewards so that every batch retains only samples carrying valid gradients. These interventions, however, incur two notable and persistent costs. (i) **Response-Level Inefficiency**: because dynamic sampling discards and regenerates responses, it intrinsically lowers sampling efficiency and slows training convergence. (ii) **Token Clipping Deficit**: our analysis establishes that the optimal clipping bound is not constant but varies from token to token as a function of the token’s prior probability, so that any uniform upper bound is inherently suboptimal. Even approaches that avoid discarding zero-gradient responses outright leave both limitations fundamentally unresolved, which motivates a method that treats the clipping bound and the advantage estimate as jointly adaptive quantities.

We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a unified reinforcement-learning method that resolves both limitations at their source through three coordinated design choices. First, we develop a **smooth advantage standardization technique that standardizes advantages across cumulative steps**, directly eliminating the vanishing-gradient problem that arises when responses sharing an identical reward are assigned zero advantage. By aggregating the reward distribution of all cumulatively generated responses with that of the current step, the technique restores useful learning signals to otherwise discarded responses and thereby enables markedly more efficient response-level exploration (see Section 2). Second, we propose a **dynamic clipping mechanism that adaptively adjusts the clipping bounds according to the old token-specific probabilities**, replacing the fixed—whether symmetric or asymmetric—clipping bounds of prior methods: tokens assigned lower prior probability are granted correspondingly wider clipping bounds, opening additional room for token-level exploration exactly where the high-entropy signal is most informative (see Section 2 and Appendix A.6). Third, DCPO computes the mean loss over the tokens within each individual response rather than over an entire batch; this Only-Token-Mean formulation preserves the relative advantage structure among responses to the same prompt and eliminates the dilution that batch-level averaging otherwise induces. The three mechanisms are complementary rather than incidental: adaptive clipping restores token-level exploration, cumulative standardization restores response-level utilization, and the response-normalized loss preserves the advantage structure that the other two depend upon.

We evaluate these mechanisms on a training corpus that merges Math-DAPO-17k (Yu et al., 2025) with levels 3–5 of the MATH dataset (Liu et al., 2025), assessing performance on four widely used mathematical reasoning benchmarks. Across four distinct models spanning both instruct and base variants, and using GRPO, DAPO, and—where available—GSPO as baselines, DCPO consistently attains the strongest results, with the largest margins concentrated on the most challenging AIME benchmarks: on AIME24-Avg@32 under 32-fold sampling, DCPO-7B scores 38.8 against 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO), while on AIME25-Avg@32, DCPO-14B reaches 19.0 against 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). These accuracy gains are inseparable from clear efficiency gains: DCPO raises the average nonzero-advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude relative to the baselines, all while delivering superior end-task performance.

For preliminary background, including the loss and advantage formulations of prior algorithms such as GRPO, DAPO, and GSPO, we refer the reader to Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Clipping the probability ratio is a standard mechanism in reinforcement learning for stabilizing policy updates and forestalling the large, destabilizing parameter shifts that off-policy optimization can induce. Its role becomes especially critical under importance sampling, the mechanism through which off-policy correction is realized. When importance sampling reweights samples by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, it yields an unbiased estimate of $\mathbb{E}_{q(x)}[f(x) \frac{p(x)}{q(x)}]$; unbiasedness, however, comes at the cost of potentially severe variance inflation, as characterized in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To curb this variance inflation, prior works (e.g., Schulman et al. (2017)) constrain the probability ratio to a fixed clipping bound governed by the hyperparameter ϵ , as stated in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping bound, however, systematically constrains the model’s exploration of the diversity space, and does so most acutely in the very regions where the old policy assigns low probability mass. In those regions the absolute admissible bound on decreasing probabilities is inherently narrow, which suppresses the capacity for meaningful policy updates precisely where informative, low-probability tokens reside. The asymmetric clipping bounds adopted by DAPO relax this constraint only partially and leave the underlying limitation intact. GSPO fares no better: by discarding entire responses whenever the sequence-level importance ratio exceeds its asymmetric bounds, it simultaneously wastes informative tokens and retains high-variance ones—a combination that destabilizes training rather than stabilizing it.

Controlling the bias–variance trade-off therefore calls for a more principled alternative, one that makes the admissible interval depend explicitly on the token probability rather than holding it fixed, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term by taking its maximum with zero yields a closed-form admissible interval for $r(x)$. The result is a family of **dynamic-adaptive clipping bounds** that vary with the old probability $q(x)$ and are parameterized by the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive clipping scheme systematically enlarges the admissible policy-update interval in low-probability regions, thereby restoring the exploration capacity that fixed bounds suppress while leaving the tight interval for high-confidence tokens unchanged. To guard against the instabilities that arbitrarily wide bounds could introduce, and following the dual-clipping method (Ye et al., 2020), we cap the maximum ratio at 10 for both positive and negative advantages, which prevents excessively large importance weights from overtraining the policy. The complete derivation of these bounds from the importance-sampling framework, together with a detailed analysis of their behavior, is provided in Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In prior methods such as GRPO, DAPO, and GSPO, the advantage $\hat{A}_{new,j}^i$ of the j -th response is obtained by standardizing only the rewards $R_{j=1,\dots,G}^i$ collected for the same prompt at the i -th step, as given in Equation (11) of Appendix A.3. This step-local standardization induces two distinct failure modes. First, whenever sampling randomness drives all rewards at a given step to a common value, the standardized advantage collapses identically to zero, and the prompt is excluded from parameter updates even though its responses may encode valuable differences in reasoning trajectories. Second, the randomness inherent in high-entropy sampling can yield sharply skewed label counts, producing large step-to-step fluctuations in the standardized advantage—including outright sign reversals—that destabilize training.

Recent evidence (Yue et al., 2025) establishes that RLVR refines a policy principally by adjusting the likelihood of correct samples rather than by fundamentally reshaping the underlying distribution. This finding licenses a key modeling choice: the reward distribution of responses to a given prompt can be treated as draws from a single global distribution that persists throughout training. That premise directly motivates a *cumulative* standardization, defined in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics μ_{total}^i and σ_{total}^i are computed over all responses generated so far for the same prompt.

To temper the residual fluctuations of both the step-specific standardization ($\hat{A}_{new,j}^i$) and the cumulative standardization ($\hat{A}_{total,j}^i$), we introduce two smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, each defined as a convex combination of the two standardization schemes whose step-dependent weights evolve with the training step i , as given in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

As training progresses, $\hat{S}A_{new,j}^i$ shifts weight toward the current distribution, encoding a preference for adapting to the latest policy outputs, whereas $\hat{S}A_{total,j}^i$ shifts weight toward the cumulative distribution, treating the current step’s deviation as a fine-grained adjustment within the broader distribution. The final advantage $\hat{A}_j^i$ then selects, between these two smoothed estimates, the one of smaller absolute magnitude, thereby attenuating whichever source of fluctuation—cumulative or step-specific—momentarily dominates and, in doing so, preserving training stability, as defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

A direct and decisive consequence of this construction is that, once a prompt has entered optimization, every response it generates in subsequent steps continues to contribute to the update irrespective of whether its step-local advantage vanishes. In particular, when identical rewards occur within a step, the affected responses still update the model through the residual advantage $\frac{1}{i} \hat{A}_{total,j}^i$, so that useful learning signals are retained rather than discarded and response-level data efficiency improves accordingly.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)** (the loss in Equation (13), used by DAPO) equalizes the contribution of every token, but precisely in doing so it distorts the relative relationship between responses of differing lengths. Consider two responses to the same prompt: response A with advantage 1 and length 500, and response B with advantage 0.5 and length 1500. Under TLM, A is weighted by $\frac{500}{500+1500} = 0.25$ and B by $\frac{1500}{500+1500} = 0.75$, so that their total loss contributions become $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$, respectively. The lower-advantage response B thus exerts the greater influence (0.375 versus 0.25)—a length-driven inversion of the intended advantage ordering.

The **Sequence-Level Mean loss (SLM)** (the loss in Equation (12), used by GRPO) exhibits a complementary defect: by averaging across the batch it attenuates the magnitude of each advantage, effectively scaling it down by a factor of G , and as the batch size grows this attenuation progressively erodes the correlations that the original advantage standardization was designed to encode. Since standardized advantages already carry well-defined relative relationships across responses, averaging over the $|o_i|$ tokens within a single response both preserves those relationships and distributes importance uniformly across that response’s tokens. We therefore average only over the tokens within each response and deliberately omit the batch-level average across responses—a formulation we term the **Only Token Mean loss (OTM)**, given in Equation (8).

$$\mathcal{T}_{DCPO}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

Following DAPO, the DCPO loss omits the KL-divergence constraint, and its lower and upper clipping bounds are instantiated directly by the dynamic-adaptive bounds derived in Section 2, so that the three mechanisms operate within a single, coherent objective.

4 EXPERIMENTAL SETUP

Models and Datasets To probe generality across both model scale and model type, we experiment with four models that jointly span instruct and base variants as well as general-purpose and math-specialized pretraining: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). Training draws on a merged corpus of approximately 25k mathematical problems that combines the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3–5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). Every problem is formatted with the Qwen-Math template (see Appendix A.8) for both training and evaluation, which enforces a consistent prompting protocol throughout and isolates the optimization method as the sole source of variation.

Evaluation We evaluate on four widely used mathematical reasoning benchmarks that together span a broad difficulty spectrum: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, we report Avg@1 alone, since its larger size (500 problems) already yields a stable estimate. For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) benchmarks, we report both Avg@1 and Avg@32, so that greedy and sampling-based behavior are each assessed under a robust and comprehensive protocol.

- **Avg@1**: the standard accuracy under greedy decoding, which quantifies the quality of the model’s single most probable prediction.
- **Avg@32**: the mean accuracy over 32 responses sampled per problem at temperature 1.0 and top-p 1.0, which quantifies the robustness and stability of the trained policy distribution under stochastic decoding.

Baseline Settings We adopt GRPO and DAPO as baselines throughout, and additionally include GSPO on Qwen2.5-Math-7B and Qwen2.5-14B. The remaining configuration—comprising the per-method clipping thresholds ϵ_* , the codebase, and the common training hyperparameters—is specified in full in Appendix A.9, and all methods are trained under identical data, prompting, and hyperparameter budgets to ensure a controlled comparison.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 reports the performance of GRPO, DAPO, GSPO, and our proposed DCPO on the four mathematical reasoning benchmarks—MATH500, AMC23, AIME24, and AIME25—under both greedy decoding (Avg@1) and sampling-based decoding (Avg@32), which jointly characterize each method’s accuracy, robustness, and generalization. Across every model scale from 1.5B to 14B, and across both instruct and base variants, DCPO attains the strongest average performance, matching or surpassing GRPO, DAPO, and—where available—GSPO in every configuration. This uniformity across model capacities, pretraining regimes, and decoding conditions establishes DCPO as a general-purpose policy-optimization method for reinforcement learning of LLMs rather than an approach tuned to a particular benchmark.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO achieves the highest average scores (47.2/32.8), exceeding both GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), surpassing GRPO (+1.3/+1.7) and remaining competitive with DAPO, improving Avg@1 by +1.0 while trailing by a marginal -0.4 on Avg@32. On Qwen2.5-Math-7B, DCPO attains (57.1/45.2), outperforming GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On the largest model tested, Qwen2.5-14B, DCPO delivers (53.2/39.0), with substantial gains over GRPO (+6.5/+7.7) and consistent advantages over DAPO (+1.3/+0.1) and GSPO (+4.2/+5.5) on Avg@1 and Avg@32. Taken together, these results demonstrate that DCPO improves reliably across the full range of model capacities examined and aligns the policy more effectively to the reward signal under both greedy and stochastic decoding.

Robustness on AIME under Sampling (Avg@32) DCPO’s most pronounced advantages emerge on AIME24 and AIME25, the high-difficulty competition benchmarks that most severely stress a model’s reasoning ability. Under Avg@32—which measures the capacity to produce correct responses under high-entropy sampling—DCPO exhibits markedly stronger robustness at the 7B and

Table 1: Performance across the four mathematical reasoning benchmarks (MATH500, AMC23, AIME24, AIME25), reported per model. Avg@1 is measured under greedy decoding and Avg@32 is averaged over 32 responses sampled at temperature 1.0 and top-p 1.0. Boldface marks the best value in each column, and Δ_{baseline} reports DCPO’s margin over the corresponding baseline, with red denoting an improvement. Across all four models and both decoding regimes, DCPO attains the strongest average score.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

14B scales: on AIME24-Avg@32, DCPO-7B achieves 38.8, ahead of GRPO (32.1), DAPO (31.6), and GSPO (34.9), and on AIME25-Avg@32, DCPO-14B reaches 19.0, ahead of GRPO (10.5), DAPO (15.3), and GSPO (9.9). Because the sampling metric rewards the diversity and reliability of correct generations rather than a single greedy output, these margins indicate that DCPO improves not merely average-case accuracy but also the breadth of correct reasoning trajectories the policy can realize. This joint optimization of reward fidelity and exploration under high-entropy decoding is exactly the property demanded by settings in which many plausible outputs are sampled per problem.

5.2 THE TOKEN CLIPPING RATIO (TCR)

We define the **Token Clipping Ratio (TCR)** as the proportion of tokens that clipping excludes from the policy-gradient update during back-propagation; a lower *TCR* therefore corresponds to a larger fraction of the generated sequence contributing to learning, and thus to more efficient use of each sampled response. As Figure 1 shows, the *TCR* behaves in sharply different ways across the

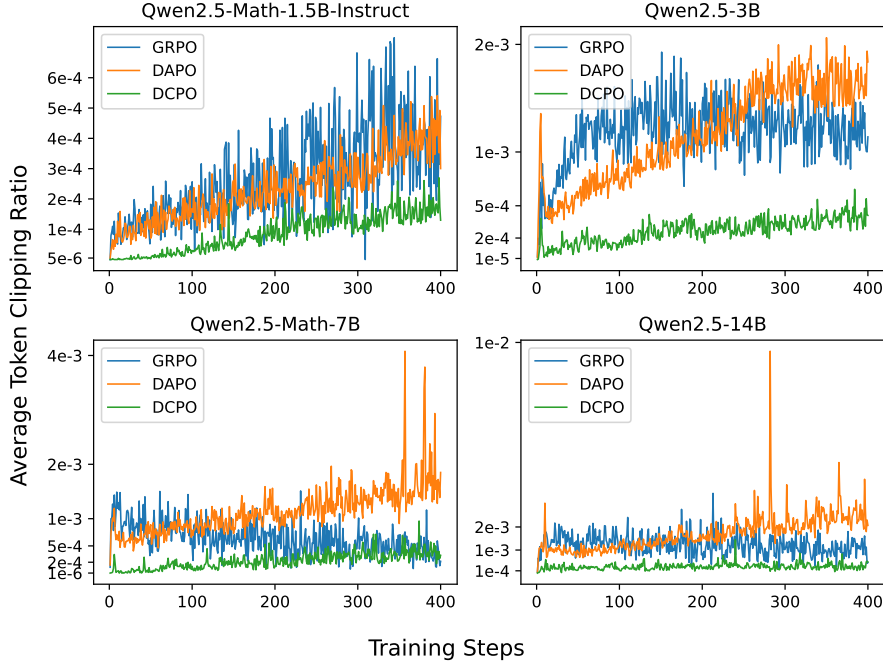


Figure 1: Token Clipping Ratio (TCR) over training for each method across the four models. Throughout training and independent of model size, DCPO sustains a TCR that is substantially lower, and markedly more stable, than that of both GRPO and DAPO.

reinforcement-learning methods considered.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

Under GRPO, the TCR follows model-size-dependent trajectories: it rises with training steps for the smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) yet gradually declines for the larger ones (Qwen2.5-Math-7B and Qwen2.5-14B). DAPO, by contrast, exhibits a monotonically increasing TCR across all models, so that as training proceeds an ever-larger share of policy updates is computed from partial or truncated responses. Both GRPO and DAPO, moreover, display pronounced fluctuations and occasional anomalous spikes in the TCR , which inject instability into optimization and can precipitate entropy collapse. GSPO is more wasteful still: its average TCR exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B—far above that of the token-level clipping methods—and consequently discards a correspondingly larger volume of informative tokens.

DCPO, in stark contrast, sustains a stable and efficient training regime. Independent of model size or training stage, its TCR remains nearly constant and stays an order of magnitude below that of GRPO and DAPO. This uniformly low and stable TCR entails that DCPO discards far fewer tokens, allowing a greater share of complete response sequences to inform each policy update and thereby opening additional room for the model to explore rare-token diversity. The net effect is a clear improvement in sample efficiency over the baselines, achieved while keeping outlier variation tightly bounded.

Taking Qwen2.5-Math-7B as a representative case, the token distribution concentrates rapidly on high-confidence tokens: after roughly 60 steps about 95% of generated tokens satisfy $q(x) > 0.9$, a fraction that rises to about 97% by roughly 100 steps and continues climbing thereafter. As illustrated in Figure 4d, once the old probability satisfies $q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$, the new probability $p(x)$ contributes to the update within the interval $[\frac{1}{1+\epsilon}, 1]$ and is never discarded under any clipping method. The overwhelming majority of tokens therefore reside in a region that is inherently immune to clipping, which accounts for the low overall TCR observed in practice. Conversely, Figure 1

shows the *TCR* rising during the later phases of most runs, signaling that a growing proportion of the remaining low-probability tokens is being discarded—precisely the tokens whose loss most severely restricts the model’s ability to explore rare-token diversity.

Prior work (Wang et al., 2025) establishes that **high-entropy tokens**—which are exactly the low-probability tokens—are the primary drivers of the emergence of reasoning capabilities in LLMs. Because entropy and probability are inversely related, rarer tokens carry proportionally more information; granting such tokens a wider admissible update range therefore enlarges the model’s effective exploration space and directly facilitates the acquisition of diverse reasoning behaviors.

These empirical and theoretical observations jointly motivate our **dynamic-adaptive clipping** scheme, which relaxes the clipping constraint for tokens with $q(x) < \frac{1}{1+\epsilon}$ while retaining the safe interval for high-confidence tokens. By adaptively widening the update interval for the high-entropy tail of the distribution, the scheme restores the model’s capacity to learn from rare, informative tokens and, in consequence, strengthens its reasoning ability—the mechanism that the low and stable *TCR* in Figure 1 makes empirically visible.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a response contributes to a policy update only when it carries a nonzero gradient. We quantify how effectively this condition is met through the **Response Utilization Ratio** (*RUR*), defined as the percentage of generated responses with nonzero advantage that actually participate in the policy update; a higher *RUR* thus signifies more efficient exploitation of the generated data.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

As Table 2 summarizes, the average *RUR* under GRPO is strikingly low—only 44.6% averaged across the four models—so that more than half of all generated responses are discarded, a substantial inefficiency in the exploitation of exploratory diversity. DCPO, by contrast, sustains an average *RUR* of roughly 70% after the first epoch, an absolute gain of 28% and a relative improvement of 64% over GRPO, and it does so consistently across all four models.

Table 2: Average *RUR* over 400 training steps, reported per model. DCPO raises response utilization to roughly 70% on every model, well above both GRPO and GSPO (“-” denotes settings in which GSPO was not evaluated).

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that GRPO’s *RUR* degrades sharply over the course of training, independent of model size or type, falling from above 90% to below 50%; on Qwen2.5-Math-7B it collapses to as low as 30% after roughly 200 steps. This decline is the direct signature of GRPO’s step-local advantage standardization, which generates a large and growing fraction of zero gradients and thereby renders exploration increasingly inefficient in the later stages of training. DCPO, driven by smooth advantage standardization, instead maintains a consistently high *RUR*: after the first epoch it stabilizes around 70% and continues to rise as training proceeds.

The *RUR* exceeds 90% during the first few steps but, from the second epoch onward, settles around 70% rather than reaching 100%. Examining precisely why it settles there, rather than treating the gap as residual inefficiency, illuminates the value of the SAS mechanism.

First, the initial *RUR* above 90% is largely an artifact of early-stage outputs that violate the required formatting, incur negative rewards, and thereby inflate the metric. That these outputs are quickly corrected indicates that the models rapidly internalize the fixed answer patterns, which is precisely why subsequent training should redirect emphasis toward exploring genuine reasoning capabilities. Second, our analysis identifies two principal sources of persistently zero advantage that hold the *RUR* near 70% rather than at 100%. (i) Nearly half of the problems are easy enough that the model solves them consistently; excluding these from updates is not a loss but a justified economy, since

the model’s core capabilities require no further refinement on them. (ii) The remaining consistently zero-advantage cases correspond to persistent errors, which arise for two reasons. On the one hand, some problems remain too difficult for the current model even given a correct ground-truth label. On the other hand, the DAPO-Math-17K dataset contains items whose labels were converted to mislabeled integers, introducing spurious “incorrect” targets. In both cases, discarding zero-advantage responses functions as an implicit data filter, mitigating the risk of updating the model on noisy or mislabeled data. The plateau near 70% is therefore a feature of the mechanism rather than a shortfall of it.

Because DAPO, unlike GRPO and DCPO, discards responses with identical labels from the generated data, a direct *RUR* comparison is not meaningful; we therefore gauge DAPO’s efficiency through the number of generated responses and the wall-clock time required to complete an equal number of update steps. To offset the low sampling efficiency that GRPO’s low *RUR* would otherwise cause, we set DAPO’s generation batch size to 1–4 times the training batch size and adjust it dynamically. Even so, across the four models DAPO requires 3 to 5 times more generated responses than DCPO to reach the same number of parameter updates, which at least doubles the training GPU hours while yielding no performance gain. This inefficiency compounds with scale: on Qwen2.5-14B, the fraction of responses DAPO actually uses falls below 30% after about 300 steps. Dynamic sampling thus combines low data utilization with poor training efficiency and forfeits substantial opportunity to explore model diversity.

In summary, DCPO advances decisively beyond both GRPO and DAPO: it substantially raises response utilization and training efficiency, converts the full pool of sampled responses into usable exploration signal for model diversity, and translates that efficiency directly into improved end-task performance.

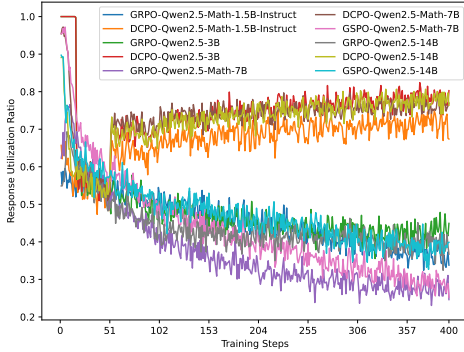


Figure 2: Evolution of the *RUR* during training. GRPO’s utilization decays sharply, whereas DCPO stabilizes near 70% and remains there for the remainder of training.

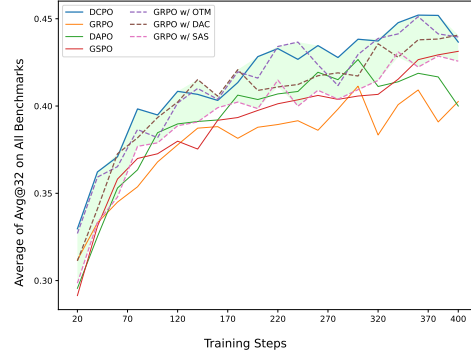


Figure 3: Component ablation on Qwen2.5-Math-7B, measured by average Avg@32 over training. Each DCPO component (OTM, SAS, DAC) is added individually to GRPO, and their combination constitutes the full DCPO; every single component surpasses GRPO, and the full method dominates all baselines.

5.4 ABLATION STUDIES

To isolate the contribution of each DCPO component, we conduct an ablation on Qwen2.5-Math-7B using **Avg@32**, the metric most sensitive to the robustness and stability of the learned policy distribution. For a controlled comparison, each variant modifies exactly one component of the baseline GRPO framework while holding all other settings fixed and removing the KL-divergence term, so that the configuration is aligned with DAPO, GSPO, and the full DCPO. The corresponding learning curves from step 20 to step 400 are reported in Figure 3.

Each of the three components, added in isolation, drives a monotone improvement in reasoning ability: every variant surpasses GRPO and matches or exceeds DAPO and GSPO. Their combination then produces a substantial aggregate gain over all baselines, confirming the effectiveness of the complete DCPO pipeline for reinforcement-learning tasks. The distinction sharpens in the later stages of training, where the baselines plateau while DCPO continues to trend upward, both at the

level of individual components and in aggregate. The ablation thus establishes that each component contributes positively to training efficiency and that their integration yields cumulative gains together with greater stability. We detail each contribution below.

(i) **Only-Token-Mean loss (OTM)**: aggregating the loss within each individual response rather than across the whole batch already lifts GRPO above the baselines, which suffer from lower data utilization and more volatile performance. (ii) **Smooth Advantage Standardization (SAS)**: replacing the stepwise advantage with the cumulative smoothed advantage markedly improves GRPO and approaches DAPO—direct evidence that aggregating reward statistics across training steps stabilizes optimization and raises final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: substituting a probability-adaptive clipping boundary for GRPO’s fixed boundary yields the largest single-component gain—DAC alone surpasses GRPO, DAPO, and GSPO, and does so more stably than OTM. (iv) **Full DCPO (OTM + SAS + DAC)**: integrating the three mechanisms delivers the best and most stable results overall, decisively outperforming GRPO, DAPO, and GSPO. Their synergy confirms that each design choice contributes a distinct and complementary benefit to data-efficient, stable reinforcement learning, jointly improving both performance and robustness for large language models.

Overall, the ablation confirms that every DCPO component contributes positively to end performance and that their combination produces substantial cumulative gains, validating the proposed mechanisms as effective and general levers for improving data efficiency and stability in reinforcement learning for LLMs.

6 CONCLUSION

This work resolves two central limitations of existing policy-optimization methods: the restricted token-level exploration imposed by fixed clipping in low-probability regions, and the low sample efficiency induced by step-local reward standardization. We introduced **Dynamic Clipping Policy Optimization (DCPO)**, a reinforcement-learning pipeline that strengthens the reasoning capabilities of large language models through two complementary innovations: (i) a **dynamically adaptive token-level clipping** mechanism that widens the update interval as a function of the old-policy probability, thereby enabling efficient exploration of rare, low-probability tokens, and (ii) a **Cumulative-Smooth Advantage Standardization** that aggregates reward statistics across optimization steps, thereby damping the reward randomness introduced by high-entropy sampling. Extensive experiments on four mathematical reasoning benchmarks across four models—spanning instruct and base variants at scales from 1.5B to 14B—substantiate these claims: DCPO-7B, for instance, attains an Avg@32 of 38.8 on the challenging AIME24 benchmark, surpassing both GRPO (32.1) and DAPO (31.6). Beyond accuracy, DCPO raises the response utilization ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude. Together these results establish DCPO as a general and effective method that markedly improves data utilization and diversity exploration in RLVR. Future work will extend DCPO to further domains, such as code generation and semantic reasoning.

REFERENCES

- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Naveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Daya Guo, Dejian Yang, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Haowei Zhang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Li, Hui Qu, J. L. Cai, Jian Liang, Jianzhong Guo, Jiaqi Ni, Jiashi Li, Jiawei Wang, Jin Chen, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, Junxiao Song, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Lei Xu, Leyi Xia, Liang Zhao, Litong Wang, Liyue Zhang, Meng Li, Miaojuan Wang, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingming Li, Ning Tian, Panpan Huang, Peiyi Wang,

- Peng Zhang, Qiancheng Wang, Qihao Zhu, Qinyu Chen, Qiushi Du, R. J. Chen, R. L. Jin, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, Runxin Xu, Ruoyu Zhang, Ruyi Chen, S. S. Li, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shaoqing Wu, Shengfeng Ye, Shengfeng Ye, Shirong Ma, Shiyu Wang, Shuang Zhou, Shuiping Yu, Shunfeng Zhou, Shuting Pan, T. Wang, Tao Yun, Tian Pei, Tianyu Sun, W. L. Xiao, Wangding Zeng, Wanjia Zhao, Wei An, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, X. Q. Li, Xiangyue Jin, Xianzu Wang, Xiao Bi, Xiaodong Liu, Xiaohan Wang, Xiaojin Shen, Xiaokang Chen, Xiaokang Zhang, Xiaosha Chen, Xiaotao Nie, Xiaowen Sun, Xiaoxiang Wang, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xingkai Yu, Xinnan Song, Xinxia Shan, Xinyi Zhou, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, Y. K. Li, Y. Q. Wang, Y. X. Wei, Y. X. Zhu, Yang Zhang, Yanhong Xu, Yanhong Xu, Yanping Huang, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Li, Yaohui Wang, Yi Yu, Yi Zheng, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Ying Tang, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yu Wu, Yuan Ou, Yuchen Zhu, Yudian Wang, Yue Gong, Yuheng Zou, Yujia He, Yukun Zha, Yunfan Xiong, Yunxian Ma, Yuting Yan, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Z. F. Wu, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhen Huang, Zhen Zhang, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhibin Gou, Zhicheng Ma, Zhigang Yan, Zhihong Shao, Zhipeng Xu, Zhiyu Wu, Zhongyu Zhang, Zhuoshu Li, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Ziyi Gao, and Zizheng Pan. Deepseek-v3 technical report, 2024. URL <https://arxiv.org/abs/2412.19437>.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021a.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *Sort*, 2(4): 0–6, 2021b.
- Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Huang, Kashif Rasul, Longhui Yu, Albert Q Jiang, Ziju Shen, et al. Numinamath: The largest public dataset in ai4maths with 860k pairs of competition math problems and solutions. *Hugging Face repository*, 13:9, 2024.
- Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding r1-zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.
- MAA. American invitational mathematics examination (AIME). Mathematics Competition Series, 2024. URL <https://maa.org/math-competitions/aime>.
- OpenAI. Introducing openai o3 and o4-mini. 2025. URL <https://openai.com/index/introducing-o3-and-o4-mini/>.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. *arXiv preprint arXiv:2409.19256*, 2024.

- Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, et al. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning. *arXiv preprint arXiv:2506.01939*, 2025.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024a.
- An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, et al. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024b.
- Deheng Ye, Zhao Liu, Mingfei Sun, Bei Shi, Peilin Zhao, Hao Wu, Hongsheng Yu, Shaojie Yang, Xipeng Wu, Qingwei Guo, et al. Mastering complex control in moba games with deep reinforcement learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pp. 6672–6679, 2020.
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- Yufeng Yuan, Qiyang Yu, Xiaochen Zuo, Ruofei Zhu, Wenyuan Xu, Jiaze Chen, Chengyi Wang, TianTian Fan, Zhengyin Du, Xiangpeng Wei, et al. Vapo: Efficient and reliable reinforcement learning for advanced reasoning tasks. *arXiv preprint arXiv:2504.05118*, 2025.
- Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025.
- Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, et al. Group sequence policy optimization. *arXiv preprint arXiv:2507.18071*, 2025.

A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In prior methods, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for token t in response j is obtained by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards over the G responses generated at the i -th step, as given in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When the rewards for a given prompt are identical, the corresponding responses receive zero advantage and contribute nothing to the update, so the sampled data are wasted.

A.3.1 GRPO

Shao et al. (2024) proposed **Group Relative Policy Optimization (GRPO)**, which samples G responses per query, assigns rewards R via a rule-based reward function, and estimates token-level advantages as in Equation (11). The model parameters are then updated according to Equation (12), the **Sequence-Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \| \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages over all tokens in a response and then over all G responses in the batch; the $\mathbb{D}_{\text{KL}}$ penalty constrains the policy against excessive deviation from a reference policy (e.g., a pre-trained model); and the fixed clipping threshold ε bounds the change in token probability ratios to prevent overly large updates. Widely adopted though it is, this batch-level averaging inherently dilutes per-response importance.

A.3.2 DAPO

Building on GRPO, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**. Observing that longer sequences exert a disproportionate influence on the GRPO gradient, they normalize by the total token count across the entire batch, which yields the **Token-Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. DAPO discards every prompt whose responses all share an identical reward and regenerates responses to restore the batch size—a procedure that, while restoring valid gradients, incurs severe data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping injects high-variance noise into gradient estimates—noise that accumulates over long sequences and is further amplified by the clipping operation itself. GSPO mitigates this by replacing token-wise clipping with sequence-wise clipping, which produces a more stable training signal at the sequence level.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

where $s_i(\theta) = \exp\left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log\left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{\text{old}}}(o_{j,t}|q)}\right)\right)$

This sequence-level criterion, however, discards more than 10% of responses that carry nonzero advantages, so its stability is purchased at the price of considerable data waste and reduced training efficiency. Moreover, while GRPO discards responses exhibiting sequence-level high variance, it still retains a large number of responses with token-level high entropy, and these continue to destabilize training.

A.4 REWARD CALCULATION

To supply a verifiable and interpretable supervision signal, we adopt a rule-based reward model that scores each generated response jointly on answer correctness and format compliance, as defined in Equation (15). Concretely, a reward of +1 is assigned when both the format and the answer are correct, 0 when the format is correct but the answer is incorrect, and −1 when the format itself is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response within the group for a given problem at the i -th training step, and the reward is determined jointly by the format compliance and answer correctness of that response.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a crucial technique in reinforcement learning for stabilizing policy optimization and preventing large, destabilizing parameter updates. Its importance is heightened whenever importance sampling is used to estimate expected values under a new policy from samples drawn by an old one, since it is precisely in this off-policy setting that unbounded ratios can inflate the estimator variance.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be re-expressed as an expectation under the old probability $q(x)$ through the importance-sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x) \frac{p(x)}{q(x)} q(x) dx = \mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be substantially inflated—a well-known difficulty of importance sampling. The true variance of $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator computed under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} [f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x) \frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

Taking the difference between the importance-sampled variance and the true variance makes the potential for inflation explicit, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To contain this discrepancy, many prior works—including PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025)—impose a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, inherently constrains the policy’s capacity for exploration, and this limitation is most damaging precisely where the old policy assigns low probability mass. In such regions the absolute bound on low probabilities is severely restricted, which suppresses the potential for meaningful policy updates. Asymmetric clipping bounds, as introduced by DAPO, address only the symmetry of the constraint, so that the fundamental limitation of a non-adaptive bound persists.

Moreover, recent studies such as Yue et al. (2025) indicate that the reasoning paths exploited by RLVR models already reside within the sampling distribution of the base model—a conclusion reinforced by the minimal parameter shift observed during off-policy batch updates, which manifests as a small fraction of clipped parameters. Motivated by this stability, we propose a more principled alternative that constrains the probability ratio $r(x)$ through a dynamic-adaptive mechanism, incorporating the probability directly into the restriction as in Equation (21) and thereby replacing the conventional fixed clipping of Equation (20).

$$|(\frac{p(x)}{q(x)} - 1)p(x)| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality and expanding yields the equivalent expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to be mathematically valid and meaningful, it must respect the fundamental constraints on probability distributions and their ratios. As stated in Equation (23), the probabilities $p(x)$ and $q(x)$ both lie in $[0, 1]$; consequently the probability ratio $r(x)$ is non-negative, as is the clipping hyperparameter ϵ .

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To guarantee both the validity and the practical applicability of the clipping bound derived from Equation (22), we specify the necessary conditions on its bounds, as set out in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with Equation (24), the probability ratio $r(x)$ must satisfy the following case-dependent conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we restrict attention to the overlapping region centered on the ratio value 1, since this region minimizes the discrepancy between the two conditions and preserves consistency with the conventional fixed bound of GRPO; the resulting validity conditions are:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we guarantee that the radicand remains non-negative by taking its maximum with zero, e.g. $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This finally yields the conditions below, which constitute the **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower bounds in Inequality 27, we assign distinct thresholds ϵ_{high} and ϵ_{low} so that the dynamic-adaptive bound coincides exactly with the GRPO bound at the designated reference points.

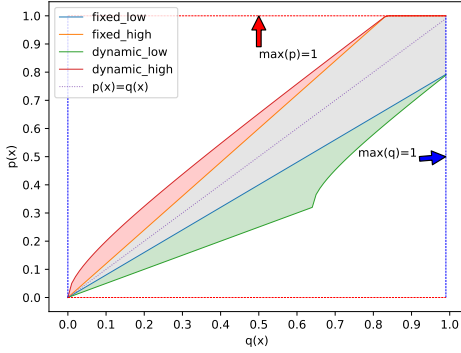
A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To make our dynamic-adaptive clipping bounds coincide with the fixed symmetric bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we obtain the specific values of ϵ_{low} and ϵ_{high} by solving the equations in Equation (28).

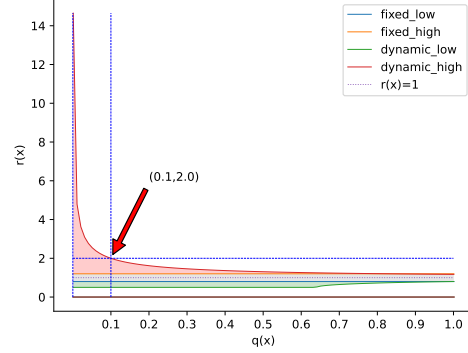
$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases} \quad (28)$$

$$\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases}$$

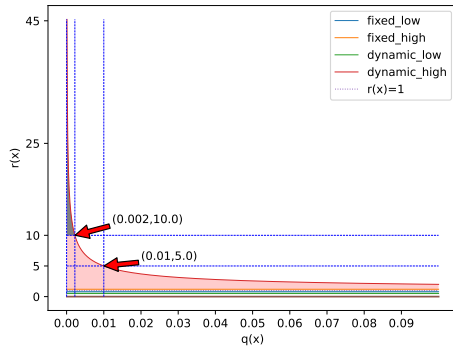
Under the standard practice of prior work (e.g., GRPO), where the fixed clipping threshold is $\epsilon = 0.2$, solving these equations gives our hyperparameters as approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Although $\epsilon_{low} \neq \epsilon_{high}$, this asymmetry serves a fundamentally different purpose from the asymmetric fixed clipping of DAPO: rather than statically relaxing the upper limit, it calibrates a probability-dependent bound so as to align with the fixed bound at the reference points.



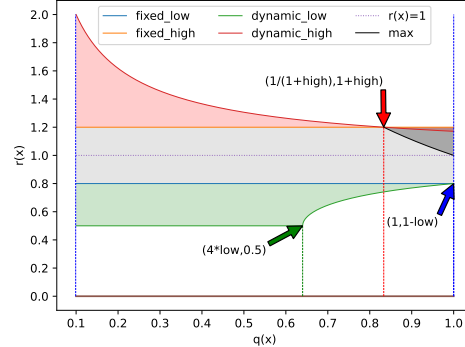
(a) Not Clipped $p(x)$ regions.



(b) $r(x)$ bounds ($q(x) \in [0, 1]$).



(c) $r(x)$ bounds ($q(x) \in [0, 0.1]$).



(d) $r(x)$ bounds ($q(x) \in [0.1, 1]$).

Figure 4: Comparison of clipping bounds as a function of the old probability $q(x)$: fixed clipping ($\epsilon = 0.2$) versus dynamic-adaptive clipping ($\epsilon_{low} = 0.16, \epsilon_{high} = 0.2$). As $q(x)$ decreases, the dynamic bound widens, expanding the admissible update interval for rare tokens, while for high-probability tokens it coincides with the fixed bound—so that the two schemes agree exactly where clipping is benign and diverge precisely where exploration is most needed.

Dynamic versus fixed clipping: Figure 4 contrasts the behavior of the fixed clipping scheme used in GRPO ($[1 - \epsilon, 1 + \epsilon]$) with that of our probability-aware adaptive clipping. For a token with old

probability $q(x)$, the fixed scheme supplies a constant relative width 2ε , so that as $q(x)$ decreases the absolute exploration space it permits shrinks linearly (Figure 4a). The dynamic scheme, by contrast, widens its relative interval inversely with $q(x)$, so that the admissible exploration space remains approximately constant across the entire probability range.

Effect of decreasing $q(x)$: As $q(x)$ becomes small, the fixed interval $[1 - \varepsilon, 1 + \varepsilon]$ remains independent of $q(x)$, so the model gains no additional room to explore low-probability tokens; the dynamic-adaptive bound, in contrast, expands rapidly (Figure 4b), granting a larger relative space to rare tokens—exactly the behavior required for efficient RL.

Behavior for extremely low probabilities: Figure 4c depicts the clipping interval for $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound on $r(x)$ grows rapidly, at a rate inversely proportional to the square root of $q(x)$, whereas the fixed bound stays constant. Following the dual-clipping strategy (Ye et al., 2020), we impose a hard ceiling $r_{\max} = 10$ to forestall the attendant instabilities (gradient vanishing, explosion, etc.). The gray region in Figure 4c, corresponding to approximately $q(x) \leq 2 \times 10^{-3}$, marks where the adaptive upper limit is clamped to $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\varepsilon$ the dynamic-adaptive scheme yields $dynamic_{low} = 0.5$, whereas the fixed scheme retains the constant $fixed_{low} = 1 - \varepsilon = 0.8$; once $q(x)$ exceeds this threshold, the dynamic-adaptive lower bound contracts smoothly toward the fixed value. Symmetrically, the dynamic-adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\varepsilon_{high}}$ and merges with it for larger $q(x)$. The adaptive interval $[dynamic_{low}(q(x), \epsilon_{low}), dynamic_{high}(q(x), \epsilon_{high})]$ therefore delivers a probability-dependent widening of the admissible region for rare tokens while preserving the tightness of the fixed interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) comprises both a positive- and a negative-advantage term. Under the minimum operation, the ratio $r(x)$ for a positive advantage is clipped to the interval $[0, dynamic_{high}(q(x), \epsilon_{high})]$, whereas for a negative advantage it is clipped to $[dynamic_{low}(q(x), \epsilon_{low}), \infty)$. Because an excessively large $r(x)$ can destabilize training, we cap the ratio at $r_{\max} = 10$ in both cases; this safeguard, following Ye et al. (2020), ensures that the adaptive limits never produce excessively high importance weights.

In short, for both positive and negative advantages, dynamic-adaptive clipping affords the model strictly more room to explore diversity for low-probability rare tokens than fixed clipping does, while remaining indistinguishable from it for high-confidence tokens.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i index the iteration at which prompts are used to generate responses and update the model parameters. We define the following statistics for reward normalization.

The mean rewards of the responses are given by the family of functions in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total} = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

The corresponding variances of the response rewards are given by the family of functions in Equation (30):

$$\begin{cases}
\sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\
\sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{old}^k)^2} \\
\sigma_{total}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^k)^2} \\
= \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)}
\end{cases} \quad (30)$$

To reduce resource usage, we compute σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29) through the equivalent recursions expressed in terms of μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i , rather than by direct accumulation over the original rewards; the two formulations are algebraically identical while the recursive form avoids storing the full reward history.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments in this paper employ the following prompting template, adapted from the Qwen-Math template (Yang et al., 2024b) and applied uniformly to training and evaluation.

Qwen-Math Template

```

<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant

```

A.9 TRAINING SETTING

We configure the clipping thresholds per method as follows: a symmetric threshold $\epsilon = 0.2$ for GRPO; lower/upper thresholds (0.2, 0.28) for DAPO, augmented with a 512-token soft-punish cache and a maximum probability ratio of 10 for negative advantage; lower/upper thresholds ($3e-4$, $4e-4$) for GSPO; and low/high thresholds (0.16, 0.2) for DCPO. The DCPO thresholds are chosen so as to preserve exactly the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and $(q(x), r(x)) = (1, 1-\epsilon)$ as GRPO (Figure 4d); unlike DAPO, we cap the maximum probability ratio at 10 for both positive and negative advantages.

Training runs for 400 steps with a generation batch size of 512 and a parameter-update mini-batch size of 32. Using temperature 1 and top-p 1, we generate $G=16$ responses per problem, with the prompt and response maximum lengths set to 1,024 and 3,072 tokens, respectively—the latter far below the 20k maximum adopted in the original DAPO, underscoring the efficiency of our setup.

Our training codebase is adapted from Verl (Sheng et al., 2024) and runs on $32 \times H20$ GPUs; unless stated otherwise, all hyperparameters match those of DAPO. For loss computation we retain the exact $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation of GRPO and DAPO rather than the approximate weighting used in Verl, which guarantees correctness across differing microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports the average Avg@1 metric, obtained by averaging the greedy-decoding (Avg@1) results over the MATH500, AMC23, AIME24, and AIME25 benchmarks, which together characterize the model’s best-prediction performance. Across all four models, DCPO

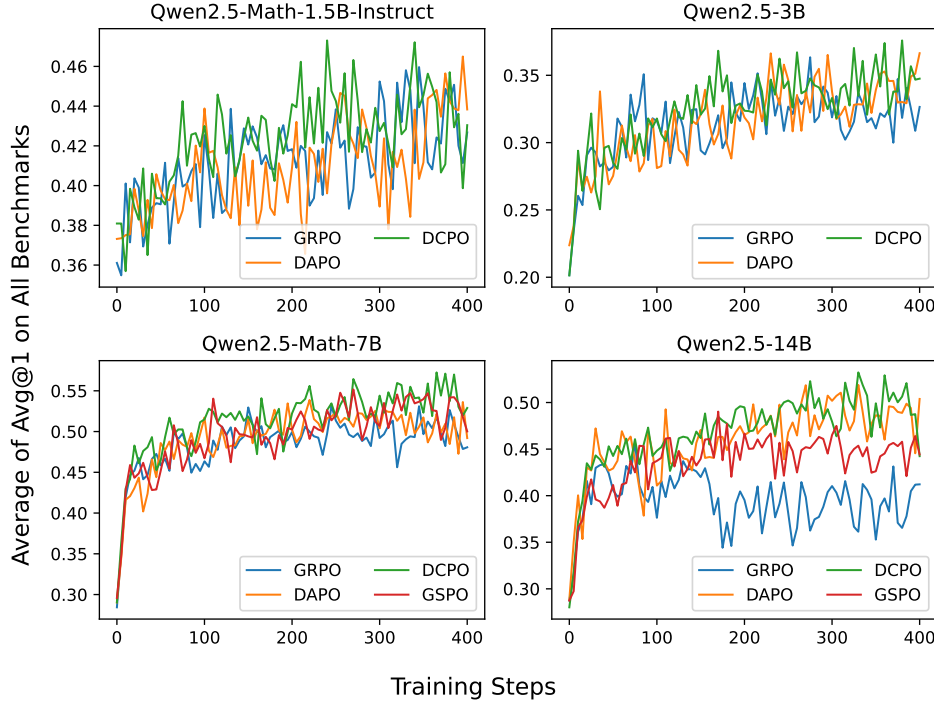


Figure 5: Avg@1 performance over training, averaged across the four benchmarks, for each model and method. DCPO consistently leads the baselines while following a markedly smoother trajectory.

consistently outperforms both GRPO and DAPO, and it also surpasses GSPO on the two models where GSPO was evaluated. Its trajectory is, moreover, markedly more stable than those of the baselines: it is free of the pronounced performance drop exhibited by GRPO and of the large fluctuations exhibited by DAPO. These trends indicate that DCPO elicits stronger reasoning capabilities over the course of training and leaves greater headroom for the model to attain still higher performance.

Avg@32 Trend Figure 6 compares the Avg@32 performance of DCPO against GRPO, DAPO, and—where available—GSPO across model sizes on the AMC23, AIME24, and AIME25 benchmarks, and it demonstrates DCPO’s exceptional robustness and stability under sampling. GRPO, by contrast, exhibits substantial fluctuation and occasional collapse: on Qwen2.5-14B it suffers a marked performance drop after approximately 170 steps. Across every evaluated model (1.5B to 14B), the DCPO trajectory trends consistently upward with minimal fluctuation, sustaining the fastest early-stage growth rate and preserving that upward trend thereafter. Relative to DAPO, DCPO furthermore attains superior performance while halving both training time and resource consumption. These results establish the effectiveness and robustness of dynamic-adaptive clipping and smooth advantage standardization in improving diversity exploration at both the response and label levels.

Entropy Trend Figure 7 shows that, after an initial warm-up, GRPO’s entropy decays sharply and then settles into a flat, low-entropy regime. This entropy collapse reflects a rapid loss of policy randomness, which reduces the diversity of sampled responses and thereby constrains subsequent policy optimization.

The behavior of DAPO and DCPO is model-dependent. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining trajectories. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), however, the trends diverge markedly: DAPO sustains the highest entropy, trending clearly upward but with large fluctuations. On Qwen2.5-Math-7B, GSPO’s entropy closely tracks that of DCPO, and their average metric performance is correspondingly comparable; on Qwen2.5-14B, by contrast, GSPO’s performance follows a fluctuating downward trend that yields poor results. DCPO instead settles at a moderate entropy level, tracing a substantially

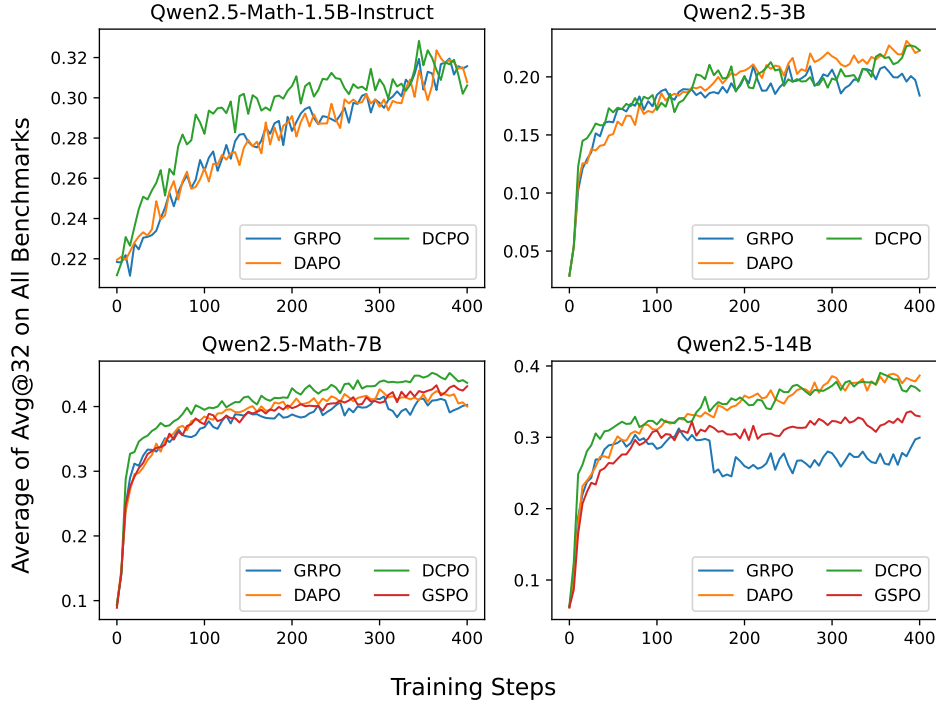


Figure 6: Avg@32 performance over training across the AMC23, AIME24, and AIME25 benchmarks, for each model and method. DCPO exhibits the fastest early-stage growth and sustains a stable upward trend under sampling.

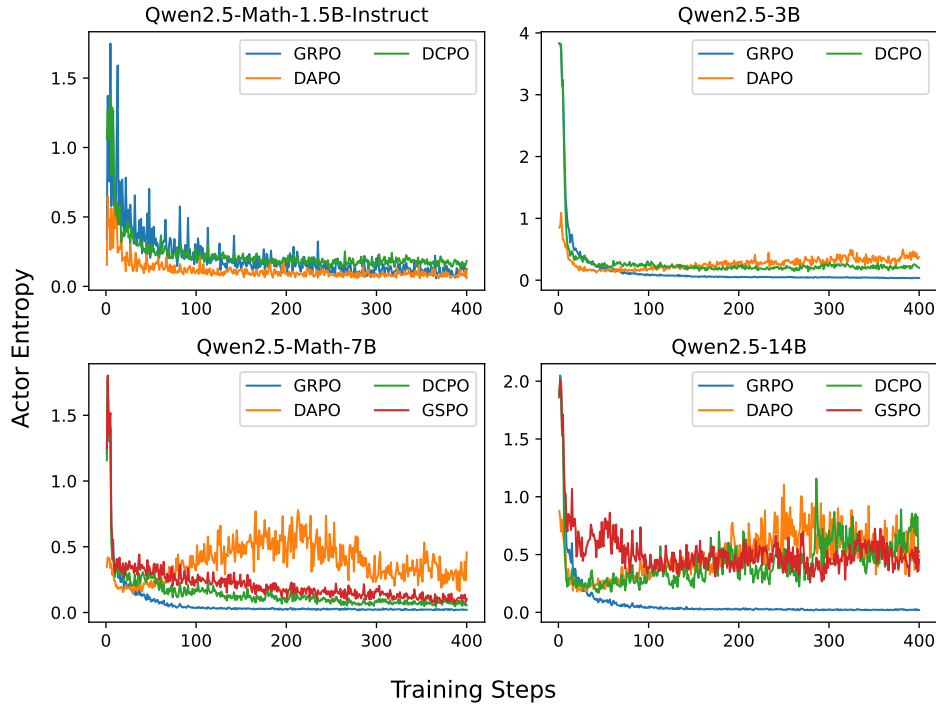


Figure 7: Policy entropy over training for each model and method. DCPO maintains entropy within a moderate, stable band, positioned between the low-entropy regime of GRPO and the high-variance regime of DAPO.

smoother curve confined to a bounded band between the low-entropy regime of GRPO and the high-variance regime of DAPO.

These observations point to a U-shaped relationship between entropy and training performance: excessively low entropy starves the learner of exploratory signal and induces premature convergence, whereas excessively high or unstable entropy undermines training stability. By holding entropy within a moderate, well-balanced range, DCPO simultaneously promotes convergence and sufficient exploration—a benefit that becomes more pronounced as model capacity increases. This entropy-regulating behavior is therefore a compelling mechanism underlying DCPO’s superior reasoning performance and its more robust policy learning across models of varying sizes.